Dunnhumby: Expanding the Candidate Score

From $\theta(c \mid S)$ to training baskets, households, and product text

Take a candidate product c and a nonempty partially filled basket S, with $c \notin S$. Let $\mathcal{I}$ be the known product catalogue, $n = |\mathcal{I}|$, and let $B \in \{0,1\}^{m \times n}$ describe the training baskets: $B_{ei} = 1$ when training basket e contains product i. Define $f_i = \sum_e B_{ei}$, its number of training-basket appearances. Each basket also has a household label $u(e)$. Product text and a product category label $\ell(i)$ are additional inputs, not quantities determined by B.

This note describes the Dunnhumby basket-completion formulation with the full revealed set S. It starts from the score, expands its terms, and ends with the rules selecting its fixed settings. All constants specify this particular formulation; their roles are explained, but their exact values are design choices unless an explicit derivation is given. We use $[x]_+ = \max(x, 0)$, $\mathbf{1}[A] = 1$ if statement A holds and zero otherwise, and stabilizers $\varepsilon_8 = 10^{-8}$, $\varepsilon_9 = 10^{-9}$, $\varepsilon_{12} = 10^{-12}$. Vector length is $\|\cdot\|_2$; a dot product $v \cdot w$ is the sum of products of corresponding coordinates. Matrix rows representing items are treated as row vectors. Sums over a set of items include each item once.

1 Start with the final score

Define $Y(c \mid S)$ as the combined evidence, $P(c \mid S)$ as a second estimate based mainly on product categories and text, $s_0 \geq 0$ as its scale, and $\kappa \geq 0$ as the strength of the second estimate. The selected settings are fixed before a new request:

$$\theta(c \mid S) = \begin{cases} Y(c \mid S), & \kappa = 0, \\ \dfrac{f_c}{f_c + \kappa} Y(c \mid S) + \dfrac{\kappa}{f_c + \kappa} s_0 P(c \mid S), & \kappa > 0. \end{cases} \tag{1.1}$$

Purpose and intuition. The assumption is that more appearances make history-based evidence more dependable. When $f_c = 0$ and $\kappa > 0$, the score is exactly $s_0 P$; when $f_c = \kappa$, the two estimates have equal weight. Frequency does not prove reliability for an individual product. Section 9 states an explicit variance assumption motivating these rational weights. We first expand Y and P.

1.1 Expand Y: exactly what changes between its two versions

Let $\mathcal{H}$ be the following ordered list of evidence sources:

$$(\text{basket}, \text{propagated}, \text{pop}, \text{cooc}, \text{household}, \text{pmi}, \text{svd}, \text{category}, \text{text}, \text{bridge}).$$

Let $w_h \geq 0$ be the selected weight of source h; a removed source has weight zero. Nonnegative weights are a restriction preventing a source's contribution from being reversed. For each source let r_h be its raw score and $z_h = \mathcal{Z}_S[r_h]$ its standardized score, with $\mathcal{Z}_S$ defined in Section 2. A second raw version r_h^0 is identical except for these two sources:

Source	$r_h^0(c \mid S)$	$r_h(c \mid S)$
basket	p_c	$p_c + \delta_c$
propagated	$(Lp)_c$	$(Lp)_c + (L\delta)_c$
All other sources	$r_h(c \mid S)$	$r_h(c \mid S)$

Here $p_c = p(c \mid S)$ is weighted training-basket support, $\delta_c = \delta(c \mid S)$ is a sum restricted to baskets containing at least two known products, and L is an item-to-item weighting matrix. Their complete definitions follow in Section 3. Define $z_h^0 = \mathcal{Z}_S[r_h^0]$. Let $\alpha \in [0,1]$ mix the two versions and $\lambda_R \geq 0$ weight the separately standardized pair-of-known-products score z_R from Section 8. Then

$$Y_0(c \mid S) = \sum_{h \in \mathcal{H}} w_h z_h^0(c \mid S), \tag{1.2}$$

$$Y_1(c \mid S) = \sum_{h \in \mathcal{H}} w_h z_h(c \mid S), \tag{1.3}$$

$$Y(c \mid S) = (1-\alpha) Y_0(c \mid S) + \alpha Y_1(c \mid S) + \lambda_R z_R(c \mid S). \tag{1.4}$$

Purpose and intuition. Only the two explicitly identified formulas change during this mixture; the other sources are held fixed by definition. This does not assert that other sources cannot represent relationships involving several products. The pair term z_R, for example, is controlled separately. Restricting α to $[0,1]$ produces an average with nonnegative weights adding to one. These are modeling choices, not requirements of every possible scoring function. Because the two raw columns are standardized separately, this is not equivalent to standardizing $p + \alpha\delta$.

1.2 Expand P: which sources enter and why

Let $\mathcal{A} = \{h : w_h > 0\}$ be the retained sources. Define the prior-source set

$$\mathcal{P} = \begin{cases} \{\text{category}, \text{text}, \text{bridge}\} \cap \mathcal{A}, & \text{if this intersection is nonempty,} \\ \{\text{category}, \text{bridge}\}, & \text{otherwise.} \end{cases} \tag{1.5}$$

For $h \in \mathcal{P}$, define $\widetilde{w}_h = w_h$ if retained, and $\widetilde{w}_h = 0.5$ otherwise. Thus the fallback rule may use a source whose weight in Y is zero. Define

$$\boxed{P(c \mid S) = \sum_{h \in \mathcal{P}} \widetilde{w}_h z_h(c \mid S) + \lambda_R z_R(c \mid S) + w_{\text{pop}} z_{\text{pop}}(c \mid S).} \tag{1.6}$$

Purpose and intuition. A product with no training appearances has a zero row in the training-only item count matrices. Its category and supplied text can still distinguish it from other products. This motivates the restricted source list. The exact list and fallback weight 0.5 are prescribed choices; the score equations do not establish that they are uniquely best. Category, bridge and agreement still partly depend on training baskets, so P is not independent of B. Selected popularity and agreement explicitly enter both estimates.

2 Expand standardization: what z_h means

For a raw source r_h, calculate its mean and population standard deviation across all catalogue products for the same revealed basket S:

$$\mu_h(S) = \frac{1}{n} \sum_{i \in \mathcal{I}} r_h(i \mid S), \tag{2.1}$$

$$\sigma_h(S) = \sqrt{\frac{1}{n} \sum_{i \in \mathcal{I}} (r_h(i \mid S) - \mu_h(S))^2}, \tag{2.2}$$

$$\mathcal{Z}_S[r_h](c) = \frac{r_h(c \mid S) - \mu_h(S)}{\sigma_h(S) + \varepsilon_9}. \tag{2.3}$$

Apply this rule separately to r_h^0 and r_R as well.

Purpose and intuition. Counts and text similarities have different units. Equation (2.3) expresses each relative to its own catalogue average and variation. A constant column becomes zero. The divisor here is n, not the sample divisor $n-1$. Known products are included in these sums and excluded only at final ranking. These are specified conventions. The hidden target set does not enter the calculation; neither do other test requests. This protects standardization from label dependence, provided the raw inputs themselves contain only information allowed at prediction time.

3 Expand the basket and propagated scores

3.1 The exact training-basket contribution

Define basket size $d_e = \sum_i B_{ei}$, known-product weight $a_s = (f_s + 5)^{-1/2}$, and basket weight $v_e = (d_e + 1)^{-1/2}$. Define weighted overlap o_e and overlap count k_e:

$$o_e(S) = v_e \sum_{s \in S} B_{es} a_s, \qquad k_e(S) = \sum_{s \in S} B_{es}, \tag{3.1}$$

$$p(c \mid S) = \sum_e B_{ec} o_e(S), \qquad \delta(c \mid S) = \sum_e B_{ec} o_e(S)\, 1.5[k_e(S) - 1]_+. \tag{3.2}$$

These give $r^0_{\text{basket}} = p$ and $r_{\text{basket}} = p + \delta$ in the table above.

Purpose and intuition. B_{ec} requires the candidate to occur in the contributing training basket. The weighted overlap adds contributions from its known products, downweighting frequent known products and large baskets. Offsets 5 and 1 keep weights finite and soften this adjustment; their precise values and the square-root powers are choices. For $k_e = 0$ or 1, the added contribution to δ is zero; for $k_e = 2$ it is $1.5 B_{ec} o_e$; for $k_e = 3$ it is $3 B_{ec} o_e$. It therefore records a specific observed overlap, not an undefined notion of group compatibility. The linear increase and multiplier 1.5 specify the chosen strength. With one known product, $\delta = 0$.

3.2 The exact item-to-item matrix L

Let $D = \text{diag}((f_i + 5)^{-1/2})$ and $E = \text{diag}((d_e + 1)^{-1})$, where diag constructs a diagonal matrix. Let off replace a matrix's diagonal by zero. Define

$$L = \text{off}(DB^\top EBD), \qquad L_{ci} = \frac{\sum_e B_{ec} B_{ei}/(d_e + 1)}{\sqrt{f_c + 5}\sqrt{f_i + 5}} \quad (c \neq i), \qquad L_{cc} = 0. \tag{3.3}$$

Writing p, δ as column vectors over all products, the other two raw scores are

$$r^0_{\text{propagated}}(c \mid S) = \sum_i L_{ci} p_i, \qquad r_{\text{propagated}}(c \mid S) = \sum_i L_{ci} p_i + \sum_i L_{ci} \delta_i. \tag{3.4}$$

Purpose and intuition. A product receives some score from other products with which it shared training baskets. The displayed entry formula defines exactly how much. Zero diagonal excludes immediate self-transfer. A product absent from all training baskets has a zero row in L.

4 Expand household evidence

Let u index households, and define the household-by-product count table

$$X_{ui} = \sum_{e : u(e) = u} B_{ei}, \qquad t_u = 1 + \sum_i X_{ui}, \qquad \widetilde{X}_{ui} = X_{ui}/\sqrt{t_u}. \tag{4.1}$$

X_{ui} counts training baskets containing product i for household u; it is not a count of purchased units. Define $C_H = \mathrm{off}(\widetilde{X}^\top \widetilde{X})$. The raw score is

$$\boxed{r_{\text{household}}(c \mid S) = \sum_{s \in S} (C_H)_{cs} (f_s + 5)^{-1/2}.} \tag{4.2}$$

For distinct products c, s, $(C_H)_{cs} = \sum_u X_{uc} X_{us} / t_u$.

Purpose and intuition. Two products can have a household connection even if they were purchased in different baskets. The division by t_u reduces the influence of households with large total counts; the extra factor on the known product reduces frequent-product influence. These weights are specified modeling choices. Every household contributes to the shared table C_H. The current shopper's household ID does not enter equation (4.2); this score is not a personalized lookup of that shopper's history. Only training baskets contribute to X.

5 Expand popularity, pair counts, and the history summary

Define $C = \mathrm{off}(B^\top B)$, $g_i = \sum_j C_{ij} + \varepsilon_9$, and $Z_C = \sum_{i,j} C_{ij}$. For a pair observed together at least once define

$$M_{ij} = [\log(C_{ij} Z_C / (g_i g_j) + \varepsilon_{12})]_+; \qquad M_{ij} = 0 \text{ if } C_{ij} = 0. \tag{5.1}$$

The raw sources are

$$r_{\text{pop}}(c \mid S) = \log(1 + f_c), \quad r_{\text{cooc}}(c \mid S) = \sum_{s \in S} C_{cs}, \quad r_{\text{pmi}}(c \mid S) = \sum_{s \in S} M_{cs}. \tag{5.2}$$

Purpose and intuition. Popularity supplies a general frequency preference; C measures direct shared-basket counts. M compares a pair count with its two overall shared-count masses g_i, g_j, which differ from the basket frequencies f_i, f_j. The logarithm compresses large ratios; the positive part discards below-expected associations rather than assigning a negative raw penalty. This is a chosen treatment of negative association.

For the summary source, set $r = \min(128, n - 2)$ and let V_r contain the leading r right singular directions of M. For an exact decomposition these give the best rank-r approximation in squared error; fitted directions may approximate them numerically. Define row normalization $\mathcal{N}(F)_i = F_i / (\|F_i\|_2 + \varepsilon_8)$ and set $U = \mathcal{N}(MV_r)$, or $U = 0$ when M is zero. Then

$$r_{\text{svd}}(c \mid S) = U_c \cdot \left(\frac{1}{|S|} \sum_{s \in S} U_s \right). \tag{5.3}$$

Purpose and intuition. The vectors summarize many pair relationships in a shorter description per product. The dot product compares the candidate with the mean known-product summary. Width 128 is a chosen dimension limit, not a data-derived requirement.

6 Expand product-text evidence and its bridge

For each catalogue product, concatenate its department, commodity, subcommodity and brand text. Use lowercased words and two-word phrases after removing English stop words, retaining at most 8,000 terms. Let n_{it} count term t in product i's text and $\mathrm{df}_t = |\{i : n_{it} > 0\}|$. Define the unnormalized text weight

$$A_{it} = \begin{cases} (1 + \log n_{it})\left[1 + \log((n+1)/(\mathrm{df}_t + 1))\right], & n_{it} > 0, \\ 0, & n_{it} = 0. \end{cases} \tag{6.1}$$

Normalize each nonzero row of A to length one, leaving zero rows zero, and call the resulting sparse table T. Let d_T be its column count, $r_T = \min(128, d_T - 1)$, and V_T its leading r_T right singular directions. Define the dense product-text table $F = \mathcal{N}(TV_T)$. This assumes at least two retained text terms, as in this dataset.

Purpose and intuition. The first factor in A_{it} compresses repeated words; the second reduces the relative influence of words appearing in many product descriptions. The short vectors summarize this table. The vocabulary and dimension limits are choices. There is no category-average subtraction in this definition of F.

Define the bridged feature table F_b and the two selectable text-related scores by

$$F_b = \mathcal{N}(F + 0.5\mathcal{N}(LF)), \tag{6.2}$$

$$r_{\text{text}}(c \mid S) = F_c \cdot \left(\frac{1}{|S|} \sum_{s \in S} F_s \right), \tag{6.3}$$

$$r_{\text{bridge}}(c \mid S) = F_{b,c} \cdot \left(\frac{1}{|S|} \sum_{s \in S} F_{b,s} \right). \tag{6.4}$$

Purpose and intuition. $(LF)_c = \sum_i L_{ci} F_i$ combines descriptions of products connected to c through training baskets. The bridge mixes that normalized combination with c's own description; 0.5 is its chosen multiplier. If $f_c = 0$, then $(LF)_c = 0$ but F_c may still distinguish c, so the bridge need not vanish. The sparse text similarity $T(T^\top x_S)$, with $(x_S)_i = \mathbf{1}[i \in S]$, is not a separately weighted source in this formulation. That exclusion is a source-menu restriction, not a statement of mathematical redundancy.

7 Expand product-category evidence

Let $\ell(i)$ be product i's subcommodity category. Let a, b index categories and define

$$A^\ell_{ea} = \sum_i B_{ei} \mathbf{1}[\ell(i) = a], \quad H = \text{off}((A^\ell)^\top A^\ell), \quad h_a = \sum_e A^\ell_{ea} + \varepsilon_9, \quad H_+ = \sum_a h_a. \tag{7.1}$$

For observed category pairs set

$$Q_{ab} = [\log(H_{ab} H_+ / (h_a h_b) + \varepsilon_{12})]_+; \qquad Q_{ab} = 0 \text{ if } H_{ab} = 0. \tag{7.2}$$

Then

$$\boxed{r_{\text{category}}(c \mid S) = \sum_{s \in S} Q_{\ell(s), \ell(c)}.} \tag{7.3}$$

Purpose and intuition. This transfers evidence between product categories even when particular product pairs have little history. Its marginal h_a counts category appearances in training baskets, rather than row sums of H. Repeated products of a category within one basket therefore contribute repeatedly to the count table. The diagonal of Q is zero: this source uses cross-category counts, with no additional penalty for repeating a category and no extra feature-cluster category source. These are choices specifying the model.

8 Expand the pair-of-known-products term

Define $V = \text{diag}((d_e + 1)^{-1/2})$ and $W = \text{off}(B^\top V B D)$, using D from Section 3. Let a known product s provide candidate support q_{cs}, and let A_c be its total support:

$$q_{cs} = W_{cs} + 0.15[F_{b,c} \cdot F_{b,s}]_+, \qquad A_c = \sum_{s \in S} q_{cs}. \tag{8.1}$$

For distinct known products s, t, define their similarity and pair weight

$$G_{st} = 0.5\mathbf{1}[\ell(s) = \ell(t)] + 0.5[F_{b,s} \cdot F_{b,t}]_+, \qquad K_{st} = [1 - G_{st}]_+, \qquad K_{ss} = 0. \tag{8.2}$$

The remaining raw source, standardized by equation (2.3), is

$$\boxed{r_R(c \mid S) = \frac{\sum_{s \in S} \sum_{t \in S} q_{cs} K_{st} q_{ct}}{A_c + 0.05}.} \tag{8.3}$$

Purpose and intuition. For $S = \{s, t\}$ with $s \neq t$, the numerator is exactly $2q_{cs}q_{ct}K_{st}$. Both known products must provide nonzero support for this contribution to be positive; K_{st} reduces the contribution when they are similar. For a larger S, any supported pair can contribute; matching all known products is not required. With one known product, $K_{ss} = 0$ makes the result zero. The denominator controls the scale of the product. The constants $0.15, 0.5, 0.05$ set chosen feature support, similarity mixing and smoothing. Here the descriptor is the bridged product-text vector F_b.

The matrix W has zero diagonal, whereas $p = B^\top VBDx_S$ retains diagonal contributions for known products. Both are evaluated over all products before the final known-product exclusion. This distinction affects standardization and propagation even though a known product cannot be returned as a suggestion.

9 Expand the settings selected before prediction

The weights, prior-source set, $\alpha, \lambda_R, s_0, \kappa$ are selected on held-out basket-completion examples and then frozen. Their search grids below are finite prescribed choices, not the only valid values or proven optimal spacings. In particular, $\alpha \in [0, 1]$ is a mathematical averaging domain, while a three-point grid is a restricted search within that domain. Exact cutoffs, tolerances and smoothing constants are part of this specified procedure.

9.1 Define the comparison measure

For held-out request j, let G_j be its nonempty hidden-product set and $\pi_j(r)$ the product at rank r after excluding S_j. Set $K_j = \min(20, n - |S_j|)$. Define

$$\mathrm{NDCG}_{20,j} = \frac{\sum_{r=1}^{K_j} \mathbf{1}[\pi_j(r) \in G_j] / \log_2(r+1)}{\sum_{r=1}^{\min(20,|G_j|)} 1/\log_2(r+1)}, \qquad \mathcal{J} = \frac{1}{v} \sum_{j=1}^{v} \mathrm{NDCG}_{20,j}, \tag{9.1}$$

where v is the number of selection requests. This rewards hidden products near the top of the list, with greater credit at earlier ranks. Final test labels do not enter selection.

9.2 Weights, prior sources and the two mixture coefficients

Starting with zero weights, visit the ordered list $\mathcal{H}$. At each source try $w_h \in \{0, 0.25, 0.5, 1, 1.5\}$ with other weights fixed, skip the all-zero vector, and accept only improvements in $\mathcal{J}(Y_0)$ exceeding 2×10^{-4}. Repeat for at most four passes, stopping if a pass changes nothing. Record that search's incumbent score as J_{coord}. Try removing bridge and then category if the loss is at most 3×10^{-4}, retaining at least one source. Absent weights become zero.

Next, visit $\alpha = 0, 0.5, 1$ as the outer loop and $\lambda_R = 0, 0.15, 0.25$ as the inner loop. Start its score incumbent at $-\infty$ and accept a candidate when its score is at least the incumbent minus 10^{-5}; after acceptance replace the incumbent with that candidate's score. A later near-tie can win, and

successive accepted near-ties can move downward slightly. This ordered rule is not simply the first global maximum. Call the final accepted score J_{mix} and retain its coefficients. Construct $\mathcal{P}$ as defined above, then carry $J_* = \max(J_{\text{coord}}, J_{\text{mix}})$ into the final comparison. This carried threshold can exceed the score of the final pruned mixture.

Purpose and intuition. These searches determine which evidence improves the held-out ranking. The exact visit order and acceptance tolerances define the result; they are not theoretical statements that one combination is globally optimal.

9.3 Prior scale and the frequency-dependent adjustment

Using the retained settings, flatten the Y and P scores across all catalogue products and selection requests. Let sd_0 be population standard deviation of this flattened table. The prior scale is fixed as

$$s_0 = \frac{\text{sd}_0(Y)}{\text{sd}_0(P) + \varepsilon_9}. \tag{9.2}$$

It can be zero. It is not recomputed from a test batch, and there is no additional unit-scale candidate in this formulation.

Purpose and intuition. This aligns the sizes of variations in the two estimates before combining them.

For each product with $f_i > 0$, define

$$x_i = 1/f_i, \qquad e_i = \frac{1}{v}\sum_{j=1}^{v}\left[Y(i \mid S_j) - s_0 P(i \mid S_j)\right]^2. \tag{9.3}$$

Let $\bar{x}, \bar{e}$ be their means over products with positive f_i. Define the line's slope $\widehat{b}$ and intercept $\widehat{a}$ by

$$\widehat{b} = \frac{\sum_i (x_i - \bar{x})(e_i - \bar{e})}{\sum_i (x_i - \bar{x})^2}, \qquad \widehat{a} = \bar{e} - \widehat{b}\bar{x}, \tag{9.4}$$

$$\kappa_* = \text{clip}\left(\frac{\max(\widehat{b}, 10^{-4})}{\max(\widehat{a}, 10^{-4})}, 0.05, 50\right). \tag{9.5}$$

Here $\text{clip}(t, l, u) = \min(u, \max(l, t))$. With fewer than two products having history, or a denominator at most 10^{-12}, the components cannot be distinguished; use $\kappa = 0$ only. Otherwise try $\kappa \in \{0, 0.05, 0.25, \kappa_*, 1\}$ in this order, starting with zero and accepting only improvements exceeding 10^{-4} over the carried score incumbent J_*. Update that incumbent after an accepted improvement.

Purpose and intuition. The assumed relation is $\mathbb{E}[e_i \mid f_i] = \sigma_0^2 + \sigma_e^2/f_i$. To state the assumption precisely, let an unobserved suitability be $\theta_i^* = s_0 P_i + \xi_i$ and let $Y_i = \theta_i^* + \eta_i$, with independent zero-mean normal errors of positive variances σ_0^2 and σ_e^2/f_i. Then, for $f_i > 0$,

$$\mathbb{E}[(Y_i - s_0 P_i)^2 \mid f_i] = \sigma_0^2 + \sigma_e^2/f_i, \qquad \mathbb{E}[\theta_i^* \mid Y_i, P_i, f_i] = \frac{f_i Y_i + \kappa s_0 P_i}{f_i + \kappa}, \quad \kappa = \sigma_e^2/\sigma_0^2. \tag{9.6}$$

This motivates both the fitted line and equation (1.1); the assumptions do not follow from counts alone. The positivity bounds and ratio clipping produce one candidate setting; held-out ranking comparisons choose the final value. Residuals compare the two score estimates, not scores with binary hidden-product labels.

10 What the training table and product inputs represent

The preceding equations assume B is fixed. In this Dunnhumby completion task it comes from baskets of 3–20 distinct products, capped at 30,000 after a fixed shuffle. Products appearing fewer

than twice in that capped pool are removed, and baskets shorter than three are dropped. Identical remaining product sets are deduplicated globally, retaining the first household association. A random 70/10/20 split supplies training, selection and test baskets; the training portion alone defines B and the household table X.

Product categories are subcommodity labels; missing product metadata receive separate missing-category labels. Product text and its vocabulary/short vectors use the known catalogue. The product-frequency filter is based on the pre-split capped pool. Thus the preparation assumes a known catalogue and does not estimate every preparation choice from B alone. The split is random, not chronological; date fields do not impose a time cutoff. These distinctions state the information available to the formulas rather than implicitly claiming a prediction task with unseen future products.

11 Substitute back to obtain the recommendation

For each candidate c, compute the raw terms from training baskets, training household counts, category labels and catalogue text. Standardize them with equation (2.3); insert them into Y and P, and then into equation (1.1). Exclude products already in S, rank the others by decreasing θ, and break exact ties by increasing product ID. For a requested positive integer K, return the highest K candidates, or all candidates if fewer than K remain. Sources with zero weight and no role in P contribute nothing. This completes the expansion of the score without introducing an additional shopper-specific scoring term.